DIACONU ANDREI – 412G

Homework Project OOP(Inventory management system)

In this program, we have a base class called Consumables, using three atributes. We have implicit constructors and constructor with parameters for the atributes. In this class, we defined a function which allows us to consume items, such as apples or potions. Each item has a price.

We also have a second class, a child class called Shop, in which we can sell items for the currency called dinars. Once we sell the items, the items will dissapear from the inventory, and we will be richer.

We have a third class, a base class called Armour. We have a specific part of **basic** equipment in the inventory and we want to upgrade it. In order to achieve that, we need to materials called **Steel** and **Crystallite**. We need 3 amount of each. If we decide to upgrade, the materials will be discarded from the inventory, and the **basic** equipment will become **shiny**.

CODE LISTING

```
#include <iostream>

using namespace std;

class Consumables {
    float price;
    string name;
    int number;

public:
    void set_price(float u_price);
    void set_name(string u_name);
    void set_number(int u_number);

    float get_price();
    string get_name();
    int get_number();

    Consumables();
    Consumables(float u_price, string u_name, int u_number);

    virtual int consume();
};

void Consumables::set_price(float u_price) {
```

```
    price = u_price;
}
void Consumables::set_name(string u_name) {
    name = u_name;
}
void Consumables::set_number(int u_number) {
    number = u_number;
}
float Consumables::get_price() {
    return price;
}
string Consumables::get_name() {
    return name;
}
int Consumables::get_number() {
    return number;
}
Consumables::Consumables() {
    cout << "Implicit constructor was called\n";
}
Consumables::Consumables(float u_price, string u_name, int u_number) {
    price = u_price;
    name = u_name;
    number = u_number;
    cout << "The constructor with parameters was called\n";
}
int Consumables::consume() {
    number = number - 2;
    cout << "I consumed two " << name << endl;
    return number;
}

class Shop : public Consumables {
public:
    using Consumables::Consumables;

    float sell(int quantity);
};

float Shop::sell(int quantity) {
    if (quantity > get_number()) {
        cout << "Not enough " << get_name() << " in stock!" << endl;
        return 0;
    }

    float total_price = quantity * get_price();
    set_number(get_number() - quantity);
    cout << "Sold " << quantity << " " << get_name() << " and gained " <<
total_price << " dinars" << endl;
    return total_price;
}

class Armour {
    string name1;
    string type;
    int number1;
public:
```

```
    void set_name(string u_name1) {
        name1=u_name1;
    }
    void set_type(string u_type) {
        type=u_type;
    }
    void set_number1(int u_number1) {
        number1=u_number1;
    }
    Armour() {
        cout<<"Implicit constructor was called\n";
    }

    string get_name1() {
        return name1;
    }
    string get_type() {
        return type;
    }
    int get_number1() {
        return number1;
    }

    Armour(string u_name1, string u_type, int u_number1) {
        name1=u_name1;
        type=u_type;
        number1=u_number1;
        cout<<"The constructor with parameters was called\n";
    }
    void upgrade() {
        if(type == "basic") {
            number1 = number1 - 3;
            type = "shiny";
        }
        cout << "I used three " << name1 << endl;
    }

};

int main() {
    Shop Object1(9.8, "apples", 10);
    Shop Object2(25, "potions", 7);

    cout << "One " << Object1.get_name() << " costs " << Object1.get_price()
<< " dinars" << endl;
    cout << "There are " << Object1.get_number() << " " << Object1.get_name()
<< " and they cost " << Object1.get_number() * Object1.get_price() << "
dinars" << endl;

    Object1.consume();
    cout << "There are " << Object1.get_number() << " " << Object1.get_name()
<< " and they cost " << Object1.get_number() * Object1.get_price() << "
dinars" << endl;

    int quantity;
    cout << "Enter the quantity of apples to sell: ";
    cin >> quantity;
```

```
    Object1.sell(quantity);

    cout << "One " << Object2.get_name() << " costs " << Object2.get_price()
<< " dinars" << endl;
    cout << "There are " << Object2.get_number() << " " << Object2.get_name()
<< " and they cost " << Object2.get_number() * Object2.get_price() << "
dinars" << endl;

    Object2.consume();
    cout << "There are " << Object2.get_number() << " " << Object2.get_name()
<< " and they cost " << Object2.get_number() * Object2.get_price() << "
dinars" << endl;

    int amount;
    cout << "Enter the amount of potions to sell: ";
    cin >> amount;

    Object2.sell(amount);

    Armour Object3("steel", "basic", 7);
    Armour Object4("crystallite", "basic", 4);

    cout << "There are " << Object3.get_number1() << " " <<
Object3.get_name1()<<endl;
    cout << "There are " << Object4.get_number1() << " " <<
Object4.get_name1()<<endl;

    cout<<"The type is " <<Object3.get_type() << endl;

    Object3.upgrade();
    Object4.upgrade();

    cout<< "Remaining steel: " << Object3.get_number1() << endl;
    cout<< "Remaining crystallite: " << Object4.get_number1() << endl;

    cout<< "The type is " <<Object3.get_type() <<endl;


    return 0;
}
```

Instances of the program running:

```
C:\Users\andre\CLionProjects\untitled16\cmake-build-debug\untitled16.exe
The constructor with parameters was called
The constructor with parameters was called
One apples costs 9.8 dinars
There are 10 apples and they cost 98 dinars
I consumed two apples
There are 8 apples and they cost 78.4 dinars
Enter the quantity of apples to sell:7
 Sold 7 apples and gained 68.6 dinars
One potions costs 25 dinars
There are 7 potions and they cost 175 dinars
I consumed two potions
There are 5 potions and they cost 125 dinars
Enter the amount of potions to sell:4
 Sold 4 potions and gained 100 dinars
The constructor with parameters was called
The constructor with parameters was called
There are 7 steel
There are 4 crystallite
The type is basic
I used three steel
I used three crystallite
Remaining steel: 4
Remaining crystallite: 1
The type is shiny

Process finished with exit code 0
```

```
C:\Users\andre\CLionProjects\untitled16\cmake-build-debug\untitled16.exe
The constructor with parameters was called
The constructor with parameters was called
One apples costs 9.8 dinars
There are 10 apples and they cost 98 dinars
I consumed two apples
There are 8 apples and they cost 78.4 dinars
Enter the quantity of apples to sell:9
 Not enough apples in stock!
One potions costs 25 dinars
There are 7 potions and they cost 175 dinars
I consumed two potions
There are 5 potions and they cost 125 dinars
Enter the amount of potions to sell:10
 Not enough potions in stock!
The constructor with parameters was called
The constructor with parameters was called
There are 7 steel
There are 4 crystallite
The type is basic
I used three steel
I used three crystallite
Remaining steel: 4
Remaining crystallite: 1
The type is shiny

Process finished with exit code 0
```

CONSUMABLES
1. Price 2. Name 3. Number
Consume()

ARMOUR
1. Name1 2. Type 3. Number1
Upgrade()

SHOP
1. Price 2. Name 3. Number
Sell(int quantity)